

VOCABULARY BUILDER

Life and Climate, a Co-Production – Key Words and Phrases • LEVEL C

The words and phrases below appear in the article *Life and Climate, a Co-Production* by Michael Marshall, a review of *The Story of Earth's Climate in 25 Discoveries* by Donald R. Prothero, published in *New Scientist* (9 March 2024). Study the table carefully, then complete the gap-fill exercise in Part 2.

PART 1 – VOCABULARY TABLE

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
truism	a statement that is so obviously true it barely needs to be said	<i>It is a truism that climate and life are intertwined, yet the details remain endlessly fascinating.</i>
intertwined	closely connected or linked so that they cannot easily be separated	<i>The histories of Earth's climate and its ecosystems are so deeply intertwined they must be told together.</i>
palaeontologist	a scientist who studies fossils and ancient forms of life	<i>Palaeontologist Donald Prothero has spent decades uncovering the connections between life and climate.</i>
recounting	narrating or telling the story of past events in detail	<i>The book takes on the hefty task of recounting four and a half billion years of Earth's history.</i>
hefty	large and requiring significant effort or capacity	<i>Summarising Earth's entire climate history in 25 chapters is a hefty task for any author.</i>
extended listicles	articles or books structured as a series of numbered entries or discoveries	<i>The reviewer notes that the book is the latest in a series of extended listicles — with all the pros and cons that implies.</i>
whistle-stop tour	a rapid journey through a subject, covering many topics briefly without great depth	<i>The 25-chapter format makes the book a whistle-stop tour through Earth's climatic past.</i>
chronologically	arranged in the order in which events happened in time	<i>Prothero tells the story chronologically, beginning with the formation of the Earth itself.</i>
extraterrestrial	originating from or existing beyond the Earth	<i>The opening chapter considers the likelihood of extraterrestrial life elsewhere in the universe.</i>
stretching the definition	expanding or bending the usual meaning of a word or concept	<i>By beginning with the Big Bang, Prothero is arguably stretching the definition of 'climate'.</i>
atmospheric oxygen	the oxygen present in the Earth's atmosphere, vital for most life	<i>The book covers how atmospheric oxygen rose dramatically due to photosynthetic microorganisms.</i>
photosynthetic	relating to the process by which plants and microorganisms use sunlight to produce energy	<i>It was photosynthetic microorganisms that first flooded the early Earth's atmosphere with oxygen.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
Snowball Earth	a period in Earth's history when the planet was almost entirely covered in ice	<i>The Snowball Earth episodes, when most of the planet froze over, are among the book's most striking chapters.</i>
greenhouse gases	gases such as CO ₂ that trap heat in the atmosphere and warm the planet	<i>The young Earth may have kept itself warm by retaining large quantities of greenhouse gases.</i>
icehouse climates	periods in Earth's history characterised by cold temperatures and extensive glaciation	<i>The book traces the switches between greenhouse and icehouse climates across geological time.</i>
mass extinctions	events in which a large proportion of Earth's species die out in a short geological period	<i>Each of the 'big five' mass extinctions gets its own chapter in Prothero's book.</i>
palaeontology	the scientific study of prehistoric life through the examination of fossils	<i>Prothero's background in palaeontology shapes his approach to describing ancient ecosystems.</i>
smothers	overwhelms or buries something so it becomes difficult to access or appreciate	<i>In his zeal to describe ecosystems in detail, Prothero sometimes smothers passages in species names.</i>
evocative imagery	language that creates vivid and powerful mental pictures	<i>The reviewer suggests the book would benefit from more evocative imagery and less technical detail.</i>
end-Cretaceous extinction	the mass extinction event approximately 66 million years ago that ended the age of dinosaurs	<i>The chapter on the end-Cretaceous extinction is described by the reviewer as a knockout.</i>
asteroid impact	the collision of a large space rock with the Earth, associated with mass extinction	<i>Prothero questions whether an asteroid impact was truly the sole cause of the end-Cretaceous extinction.</i>
sole cause	the only reason or explanation for something; no other factors involved	<i>The author argues against the idea that an asteroid impact was the sole cause of the mass extinction.</i>
tone and pace	the overall mood and speed at which a written work progresses	<i>The tone and pace of the book shifts noticeably as Prothero reaches the more recent geological periods.</i>
quick-fire guide	a short, fast-moving reference work that covers many topics rapidly	<i>For readers from unrelated specialisms, the book serves as an excellent quick-fire guide to the topic.</i>
specialism	an area of expert knowledge or professional focus	<i>Scientists from outside Prothero's specialism will likely find the book especially illuminating.</i>
apocalypse	a catastrophic event involving widespread destruction, often used figuratively	<i>The reviewer notes that in Prothero's account, there is always another apocalypse just a few pages away.</i>

PART 2 – GAP-FILL EXERCISE

Complete the sentences using the correct word or phrase from the word bank below. You may need to change the form of the word. Most sentences require more than one answer.

WORD BANK

truism	intertwined	palaeontologist	recounting
hefty	extended listicles	whistle-stop tour	chronologically
extraterrestrial	stretching the definition	atmospheric oxygen	photosynthetic
Snowball Earth	greenhouse gases	icehouse climates	mass extinctions
palaeontology	smothers	evocative imagery	end-Cretaceous extinction
asteroid impact	sole cause	tone and pace	quick-fire guide
specialism	apocalypse		

1	It is a _____ to say that life and climate are deeply _____ — and yet Prothero's book shows just how rich and surprising those connections truly are.
2	The author, _____ Donald Prothero, takes on the _____ task of _____ how life and climate shaped each other across 4.5 billion years of Earth's history.
3	The reviewer notes that the book is the latest in a series of _____ — with all the good and bad that implies — and functions as a _____ through Earth's past.
4	Prothero tells the story _____, beginning with the Big Bang — thereby arguably _____ of 'climate' just a little.
5	Early chapters cover the rise of _____ caused by _____ microorganisms, as well as the _____ episodes when most of the planet froze over.
6	The young Earth may have kept itself warm by retaining _____, since the newly formed sun was fainter than it is today.
7	The second half of the book traces the switches between _____ and _____ on Earth, accompanied by enormous rises and falls of sea level.
8	Each of the 'big five' _____ gets its own chapter, including the _____ — which the reviewer describes as a knockout.
9	Prothero questions whether an _____ was truly the _____ of the extinction, and objects to what he sees as an excessive focus on dinosaurs.
10	In his zeal to describe ecosystems in detail, Prothero sometimes _____ passages in species names — the reviewer suggests more _____ would have helped.
11	The _____ of the book shifts as Prothero moves into fifth gear and explains why extinction events are more complicated than 'big rock hits, everything dies'.

12

For scientists from unrelated _____, the book will be an excellent _____ to the topic — rich with concepts, stories, and characters.

13

The reviewer remarks that there is always another _____ just a few pages away — a testament to how dramatic Earth's climatic history truly is.

Write your answers here:

1. _____	2. _____	3. _____
4. _____	5. _____	6. _____
7. _____	8. _____	9. _____
10. _____	11. _____	12. _____
13. _____		